

江南大学 2022-2023-1 学年度《概率论与数理统计 I》期末考试卷 参考答案和评分标准

一、填空题 (每空 5 分, 共计 50 分) 1. $\overline{ABC} \cup C\overline{A}\overline{B} \cup B\overline{A}\overline{C}$ 2. 1.5% 3. $\frac{\sqrt{2}}{4}$ 4. $2n/\sum_{i=1}^n \frac{1}{x_i}$ 5. [6.92, 13.08]	二、计算题 1. 定义随机事件 A、B 分别为控制系统 A、B 能在此工况下正常工作, 则 $P(A) = 0.8, P(B) = 0.6, P(B \overline{A}) = 0.4$ (4 分) 由 $P(B \overline{A}) = \frac{P(B\overline{A})}{P(\overline{A})} = \frac{P(B) - P(AB)}{1 - P(A)} = 0.4$ 得, $P(AB) = 0.52$ (1) $P(A \cup B) = P(A) + P(B) - P(AB) = 0.88$ (4 分) (2) $P(A \overline{B}) = \frac{P(A\overline{B})}{P(\overline{B})} = \frac{P(A) - P(AB)}{1 - P(B)} = 0.7$ (4 分)	3. 1. 由 $\int_{-\infty}^{+\infty} k(1-x)^2 dx = \int_{-1}^1 k(1-x)^2 dx = 1$ 得 $k = \frac{3}{8}$ (4 分) 2. $F(x) = \int_{-\infty}^x f(t)dt = \begin{cases} 0, & x \leq -1 \\ \frac{x^3}{8} - \frac{3x^2}{8} + \frac{3x}{8} + \frac{7}{8}, & -1 < x \leq 1 \\ 1, & x > 1 \end{cases}$ (4 分) 3. 定义随机事件 A_i 为“第 i 次观测的观测值大于等于 0”, 则 $P(A_i) = \int_0^{+\infty} f(x)dx = \int_0^1 \frac{3}{8}(1-x)^2 dx = \frac{1}{8}$ $P(A_1 \cup A_2 \cup A_3) = 1 - P(\overline{A_1} \overline{A_2} \overline{A_3}) = 1 - \prod_{i=1}^3 P(\overline{A_i}) = \frac{169}{512} = 0.33$ (5 分)
6. $3\overline{X}$ 7. $U(0,2)$ $1/3$ 8. $4/5$ $2/5$	2. (1) 1) 提出假设 $H_0: \sigma_A = \sigma_B; H_1: \sigma_A \neq \sigma_B$ 选择检验统计量 $F = \frac{S_A^2}{S_B^2} \sim F(m-1, n-1)$ (3 分) 2) 对于显著性水平 $\alpha = 0.05, F_{0.975}(9,9) = \frac{1}{F_{0.025}(9,9)} = 0.249$, 拒绝域 $W = (0, F_{0.975}(9,9)] \cup [F_{0.025}(9,9), +\infty) = (0, 0.249] \cup [4.02, +\infty)$ (3 分) 3) 计算统计量, $F = \frac{12^2}{6^2} = 4$. 因为 $F \notin W$, 所以接受 H_0, 认为两机床加工的零件精度一致。 (3 分) (2) (不设统一标准, 合理解释即可) (3 分)	4. (1) $F_Z(z) = P(Z \leq z) = P(Y \leq X + z)$ $Z \leq -1$ 时, $F_Z(z) = 0$; $Z > 0$ 时, $F_Z(z) = 1$ $-1 < Z \leq 0$ 时, $F_Z(z) = \int_{-z}^1 \int_0^{x+z} 12y^2 dy dx = (1+z)^4$ $f_Z(z) = F'_Z(z) = \begin{cases} 0, & z \leq -1 \\ 4(1+z)^3, & -1 < z \leq 0 \\ 1, & z > 0 \end{cases}$ (7 分) (2) $E(X) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xyf(x,y) dx dy = \int_0^1 \int_0^x x(12y^2) dy dx = \frac{4}{5}$ $E(XY) = \frac{1}{2}$ (3 分) (3) $E(Y) = \frac{3}{5}, Cov(X,Y) = E(XY) - E(X)E(Y) = \frac{1}{50}$ (3 分)